Human vs GPT-4o-mini Cloze Predictions

1980-01-01

Introduction

Predictability is a central variable in psycholinguistic models of reading and language comprehension. It is typically operationalized as the conditional probability of a word given its preceding context and is most commonly estimated using the cloze procedure (Taylor, 1953). In this paradigm, participants are presented with sentence fragments and asked to provide a continuation; predictability is computed as the proportion of respondents who produce the target word.

Extensive research has demonstrated that predictability systematically modulates online processing. Highly predictable words are associated with shorter reading times, reduced fixation durations, and lower processing costs relative to less predictable words. Predictability effects have been observed across multiple dependent measures, including first fixation duration, gaze duration, total reading time, and regression probability. These effects are generally interpreted within probabilistic or expectation-based frameworks of comprehension, according to which readers continuously generate and update probabilistic expectations about upcoming linguistic input.

Importantly, predictability does not operate in isolation. It interacts with other lexical and contextual variables, including word frequency, word length, morphological complexity, and syntactic structure. Although frequency and predictability are often correlated, they reflect distinct sources of information: frequency indexes general lexical familiarity, whereas predictability reflects context-specific expectation. Empirical studies have shown that both variables independently contribute to variance in eye-movement measures, supporting the view that readers integrate global lexical statistics with local contextual constraints during processing.

Related Work

Jacobs et al. (2024) conducted a series of experiments explicitly designed to compare language model–generated continuations with human cloze norms collected on the same task. They performed a large-scale evaluation of cloze completion data and demonstrated that token prediction tasks are neither lexically nor semantically aligned with human responses. Although language models often retrieve plausible continuations, their rank correspondence with human cloze probabilities reveals systematic distortions: frequent human responses are under-ranked, whereas rare responses are over-ranked. Even the most probable human responses are retrieved in first position by models only in a minority of cases. Furthermore, analyses of semantic embedding spaces indicate that human and model-generated continuations tend to occupy partially distinct regions, suggesting distributional misalignment at the semantic level.

Ilia and Aziz (2024), using the Provo Corpus, compared full conditional probability distributions produced by humans and several large language models. Their results indicate that humans exhibit graded uncertainty in next-word prediction, whereas model distributions are substantially narrower and more concentrated. In terms of total variation distance, model-generated conditional probability distributions diverge systematically from human distributions, reflecting overconfidence and reduced dispersion.

In contrast, Rego, Snell, and Meeter (2024) demonstrated that large language model–derived predictability estimates may outperform traditional cloze probabilities in explaining eye-movement measures within a cognitive model of reading. Using autoregressive transformer-based models, they showed that model-based surprisal improved model fit across multiple eye-tracking metrics. Notably, LLaMA-based probability estimates reduced prediction error more effectively than cloze-based estimates in the OB1-reader framework.

Taken together, these findings raise a critical methodological question: can large language models serve as reliable substitutes for human-derived predictability norms? The evidence suggests that models may approximate certain aspects of processing difficulty, yet they diverge from human distributions at the lexical and semantic levels.

Research Question

A central question underlying this work is whether large language models can, in principle, serve as substitutes for human participants in the estimation of predictability norms. If such substitution is possible, it is necessary to determine under what conditions it may be justified and which aspects of human expectations are adequately captured by computational models.

The motivation for this inquiry is both theoretical and practical. From a methodological perspective, the collection of psycholinguistic norms requires substantial time, financial resources, and coordinated participant recruitment. Large-scale norming studies involve extensive human labor and are therefore costly to conduct and difficult to replicate across languages and domains. If language models could approximate human probability distributions with sufficient fidelity, they might provide a scalable alternative for norm generation.

However, the issue is not merely one of accuracy in next-word prediction. As emphasized in prior work, including Language Models for Cloze Task Answer Generation in Russian (2020), the crucial question concerns the extent to which model-derived probability distributions align with human expectations at different representational levels (lexical, syntactic, or semantic). A model may correctly retrieve the target word, yet distribute probability mass in a manner that diverges substantially from human response patterns.

In the present study, we therefore seek to evaluate this question within the framework of an experimentally validated dataset. Rather than constructing a new artificial benchmark, we assess model behavior using a cloze test that has already been administered to human participants. This design allows for a direct comparison between model-generated probability distributions and empirically observed human norms under identical contextual constraints.

Task and Data

A **cloze task** presents a sentence with the final word removed; respondents supply what comes next. The dataset covers **144 Russian sentence contexts**. Human responses were collected from 628 subjects (14–151 per context, avg 48), yielding 6 898 individual responses with 1–65 unique answers per context. GPT-4o-mini predictions were obtained as a ranked list of token continuations with log-probabilities, converted to probabilities via $\exp(\sum \text{logprob})$. After deduplication by stripped surface form, each context has between 6 and 13 957 distinct GPT candidates.

Metrics

Human accuracy — the proportion of human respondents who gave the exact ground-truth target word. In 62 % of contexts (89/144) no human guessed the target at all; mean accuracy across all contexts is 0.10 (median 0.00), reaching 1.00 only for the easiest items.

match@k — for a given k, the fraction of **unique** human answer types that appear anywhere in GPT's top-k predictions (unweighted).

weighted match@k — the fraction of **human response mass** covered: for each human answer found in the top-k, its probability (response frequency) is summed. Because popular answers contribute more, weighted match is consistently higher than unweighted match and is the more meaningful metric.

Results

Surface-Form Overlap

Code: `compare_preds.py` · Data: `output/comparison.csv`

Method

GPT-4o-mini predictions were sorted by descending probability and deduplicated by stripped surface form, retaining the highest-probability occurrence of each unique token per context. Human responses were likewise deduplicated by stripped surface form, with each unique answer retaining its cloze probability.

Two metrics were computed for $k \in \{1, 5, 10, 20, 100, 1000\}$:

- **match@k** — the fraction of **unique** human answer types that appear anywhere in GPT's top-k predictions (unweighted).
- **weighted match@k** — the fraction of **human response mass** covered: for each human answer found in the top-k, its probability (response frequency) is summed. Because popular answers contribute more, weighted match is consistently higher than unweighted match and is the more meaningful metric.

Results

Coverage saturates rapidly: almost all gains occur before $k = 10$, with negligible improvement beyond $k = 20$.

k	match@k	weighted match@k
1	0.064	0.180
5	0.176	0.363
10	0.208	0.398
20	0.213	0.404
100	0.216	0.406
1000	0.216	0.406

At $k = 10$ the model's top candidates already cover 40 % of human response mass on average, but only 21 % of unique answer **types** — reflecting that GPT reliably finds the most frequent human answers while missing rare or idiosyncratic ones.

In 9 contexts (e.g. *летела*, *миссионеры*, *обгорели*) GPT produces **zero overlap** with any human answer even at $k = 1000$, suggesting fundamental distributional divergence on semantically constrained or low-frequency items.

Interpretation

GPT-4o-mini's probability mass concentrates on the same words humans prefer most, but its vocabulary of plausible continuations is much narrower: the long tail of human creativity (typos aside) is largely absent from the model's distribution. The sharp saturation of match@k by $k = 10$ means that simply taking the model's top-10 candidates is nearly as informative as its entire ranked list.

Lexical Overlap (Lemma-Based)

Code: `analysis_lexical.py` · Data: `output/lexical/`

Method

The surface-form match@k metric reported earlier treats morphological variants of the same word as distinct types. In a morphologically rich language such as Russian, this penalises both humans and the model for producing different inflected forms of the same lexeme. To address this, we compute a lemma-based overlap metric.

GPT-4o-mini predictions were deduplicated by lemma: for each context, predictions were sorted by descending probability and only the highest-probability occurrence of each unique lemma was retained. Human responses were similarly aggregated at the lemma level: for each context, unique surface-form answers were first identified with their cloze probabilities, then grouped by lemma, summing probabilities across surface forms that share the same lemma.

Two metrics were computed for $k \in \{5, 10, 15, 20, 50, 100, 200\}$:

- **overlap@k** $= |L_{\text{human}} \cap L^k_{\text{GPT}}| \ / \ |L_{\text{human}}|$, where L_{human} is the set of unique human lemmas and L^k_{GPT} is the set of top-k GPT lemmas.
- **weighted overlap@k** $= \sum_{l \in L_{\text{human}} \cap L^k_{\text{GPT}}} p(l)$, where $p(l)$ is the summed cloze probability of lemma l.

Results

k	overlap@k	weighted overlap@k
5	0.188	0.386
10	0.219	0.420
15	0.225	0.425
20	0.227	0.429
50	0.232	0.434
100	0.233	0.435
200	0.235	0.436

For comparison, the corresponding surface-form metrics are reproduced below:

k	surface match@k	weighted surface match@k
5	0.176	0.363
10	0.208	0.398
20	0.213	0.404
100	0.216	0.406

Interpretation

Lemma-based overlap consistently exceeds surface-form matching at every k, confirming that a portion of apparent mismatches between human and model responses are attributable to morphological variation rather than genuine lexical divergence. At $k = 10$, lemma overlap reaches 0.219 (vs.
0.208 for surface match) and weighted lemma overlap reaches 0.420 (vs.
0.398), representing a relative improvement of roughly 5–6% in both type coverage and probability mass coverage.

However, the gains from lemmatisation are modest. The lemma-based ceiling at $k = 200$ (0.235 unweighted, 0.436 weighted) remains well below full coverage, indicating that the dominant source of human–model divergence is not inflectional variation but rather genuine lexical disagreement: humans and the model frequently propose different words altogether.

The saturation pattern mirrors that of the surface-form metric: nearly all improvement occurs before $k = 20$, with negligible gains beyond that point. This reinforces the finding that GPT-4o-mini's useful vocabulary of continuations is effectively exhausted within its top-20 predictions.

POS Distribution Overlap

Code: `analysis_pos_overlap.py` · Data: `output/pos_overlap/`

Method

To assess whether GPT-4o-mini captures the same syntactic preferences as human respondents, we compare the probability-weighted part-of-speech (POS) distributions produced by each source. For every sentence context, we construct two POS distributions:

- **Human POS distribution.** For each context, we first deduplicate human responses by surface form, retaining one row per unique answer with its cloze probability and Universal POS (UPOS) tag. We then sum the probabilities of all answers sharing the same POS tag, yielding a distribution over POS categories weighted by response frequency.
- **Model POS distribution.** Analogously, we deduplicate GPT-4o-mini predictions by surface form (keeping the highest-probability entry per word), then sum converted probabilities across predictions sharing the same UPOS tag.

We report two metrics. First, **POS overlap@1**: for each context, we identify the POS tag with the highest total weight in each distribution and check whether the human and model top-POS tags match. The mean across all 144 contexts gives the overall POS overlap@1 rate.

Second, we apply **Algorithm 1 (ranked POS intersection@k)**. For each context, POS tags are ranked by descending total weight separately for human and model distributions. For $k = 1, 2, \ldots, 5$, we compute:

$$\text{intersection@ } k = \frac{|\text{top- } k \text{ human POS} \cap \text{top- } k \text{ model POS}|}{k}$$

and average across all 144 contexts.

Results

The basic POS overlap@1 rate is **0.556**, indicating that the model's most-weighted POS tag matches the human top POS in roughly 56% of contexts.

k	mean POS intersection@k
1	0.556
2	0.542
3	0.502
4	0.484
5	0.464

Intersection@k decreases as k grows, falling from 0.556 at $k = 1$ to 0.464 at $k = 5$. This decline indicates that beyond the dominant POS category, human and model distributions diverge in their secondary syntactic preferences.

Interpretation

A POS overlap@1 of 56% shows that the model and humans agree on the dominant syntactic category roughly half the time. This is a moderate level of agreement — substantially above chance (given the number of possible UPOS tags), yet far from ceiling. Contexts where they disagree often involve the model assigning highest mass to punctuation (PUNCT) or function words (PART, ADP), while humans favour content-word categories such as NOUN or VERB.

The gradual decline of intersection@k with increasing k suggests that the syntactic diversity of human responses is only partially mirrored by the model. At $k = 5$ the model still recovers roughly 46% of the top human POS categories on average, indicating that the two distributions share a common core of syntactic expectations but diverge in the tails. These findings are consistent with the lexical overlap results: GPT-4o-mini captures the broad syntactic shape of human expectations while differing in finer distributional details.

POS Probability Correlation

Code: `analysis_pos_corr.py` · Data: `output/pos_corr/`

Method

While the POS overlap analysis (preceding section) asks whether humans and GPT-4o-mini *use* the same parts of speech, it does not test whether the *probability mass* assigned to each POS category is similar. Two complementary algorithms address this question.

Setup. For each of the 144 contexts, a POS probability vector is constructed for both humans and the model. Human responses are first deduplicated by surface form within each context, taking the first recorded cloze probability and UPOS tag for every unique answer; the probabilities are then summed by POS tag. Model predictions are deduplicated by `prediction_cleaned`, retaining the highest probability for each unique surface form, and likewise summed by POS. Missing POS categories receive a probability of zero.

Algorithm 2 (per-POS correlation). Each POS tag defines a pair of 144-element vectors — one from humans, one from the model — containing the summed probability mass that POS received in each context. Pearson and Spearman correlations are computed for each POS tag, restricted to tags that appear in at least 10 contexts.

Algorithm 3 (delta table). For each context, the cell-wise absolute difference $\delta_c[\text{POS}] = |p_{\text{human}}(c, \text{POS}) - p_{\text{model}}(c, \text{POS})|$ is computed for every POS tag. The mean delta for a context is the average of these absolute differences across all POS categories; the overall mean delta is the grand average across all 144 contexts. A mean delta of zero would indicate perfect distributional alignment.

Results

Algorithm 2 yields the following per-POS correlations (sorted by Pearson r, descending):

POS	Pearson r	Pearson p	Spearman r	Spearman p
NUM	0.94	< 0.001	0.58	< 0.001
ADP	0.84	< 0.001	0.40	< 0.001
SCONJ	0.76	< 0.001	0.40	< 0.001
CCONJ	0.71	< 0.001	0.47	< 0.001
NOUN	0.70	< 0.001	0.81	< 0.001
VERB	0.55	< 0.001	0.60	< 0.001

POS	Pearson r	Pearson p	Spearman r	Spearman p
ADJ	0.52	< 0.001	0.39	< 0.001
ADV	0.43	< 0.001	0.33	< 0.001
PRON	0.34	< 0.001	0.49	< 0.001
PART	0.30	< 0.001	0.30	< 0.001
DET	0.26	$= 0.001$	0.40	< 0.001
X	0.02	$= 0.776$	0.05	$= 0.513$
INTJ	-0.004	$= 0.965$	0.03	$= 0.760$
PROPN	-0.007	$= 0.938$	0.21	$= 0.012$
PUNCT	-0.04	$= 0.644$	0.01	$= 0.861$

Algorithm 3 yields an overall mean delta of **0.046** across all 144 contexts (range: 0.0002 – 0.112).

Interpretation

The per-POS correlations reveal a clear hierarchy. Closed-class functional categories that are syntactically constrained — NUM ($r = 0.94$), ADP ($r = 0.84$), SCONJ ($r = 0.76$), CCONJ ($r = 0.71$) — exhibit the strongest agreement between human cloze responses and model predictions. These categories are heavily determined by the preceding syntactic context, and both humans and GPT-4o-mini respond similarly to these constraints.

Open-class content words show moderate agreement: NOUN ($r = 0.70$), VERB ($r = 0.55$), ADJ ($r = 0.52$). The relatively high Spearman correlation for NOUN ($\rho = 0.81$) suggests strong rank-order agreement even when absolute probability magnitudes differ.

The weakest correlations are observed for categories that are either rare or context-insensitive: PUNCT ($r = -0.04$), PROPN ($r = -0.007$), INTJ ($r = -0.004$), and X ($r = 0.02$). The model's tendency to assign substantial probability to punctuation tokens (which rarely appear in human cloze responses) drives the near-zero or negative PUNCT correlation.

The overall mean delta of 0.046 from Algorithm 3 indicates that, on average, each POS category's probability differs between humans and the model by about 4.6 percentage points. While this confirms that human and model POS distributions are broadly aligned at the category level, the non-trivial per-context range (up to 0.112) shows that alignment varies considerably across sentence contexts.

Entropy-Stratified Analysis

Code: `analysis_entropy.py` · Data: `output/entropy/`

Method

To assess how distributional uncertainty in human responses relates to human–model agreement, we compute Shannon entropy for each of the 144 sentence contexts and stratify the overlap analysis by entropy quartile.

Human entropy. For each context c, let $p_1, ..., p_m$ be the cloze probabilities of the m distinct answers given by respondents (these proportions sum to 1 within each context). The Shannon entropy is

$$H_{\text{human}}(c) = -\sum_{i=1}^{m} p_i \ln p_i$$

Low entropy indicates high agreement among respondents (one dominant answer); high entropy indicates a dispersed response distribution.

Model entropy. For GPT-4o-mini, the API returns an unnormalized probability distribution over multi-token continuations: the sum of `probability_converted` per context ranges from 0.13 to 2.00 (median 0.51), with 7 of 144 contexts exceeding 1.0.[1] We compute entropy over the raw (non-renormalized) probabilities:

$$H_{\text{model}}(c) = -\sum_{j=1}^{n} q_j \ln q_j$$

where q_j is `probability_converted` for the j-th unique prediction. Because the distribution is neither complete nor guaranteed to sum to 1, H_{model} is not directly comparable in magnitude to H_{human}, but it still captures how concentrated or spread the model's probability mass is within the observed portion.

Quartile stratification. We rank all 144 contexts by H_{human} and split them into four quartiles using `pandas.qcut` (Q1 = lowest entropy, Q4 = highest). Within each quartile we report the mean lemma-based overlap@K and weighted overlap@K at $K \in \{5, 10, 20, 50, 100\}$, following the same lemma-matching procedure described in the lexical overlap analysis.

Results

Across all 144 contexts, the mean human entropy is $H_{\text{human}} = 2.25$ nats and the mean model entropy is $H_{\text{model}} = 1.00$ nats. The model's partial distribution is substantially more concentrated than the human response distribution.

Quartile	Mean H_{human}	Mean H_{model}	overlap@10	weighted overlap@10
Q1 (low)	1.15	0.93	0.379	0.747
Q2	2.05	1.11	0.256	0.451
Q3	2.55	1.05	0.145	0.261
Q4 (high)	3.25	0.90	0.096	0.222

The pattern is monotonic: as human uncertainty increases from Q1 to Q4, lemma overlap@10 drops from 37.9% to 9.6%, and weighted overlap@10 drops from 74.7% to 22.2%. A similar trend holds across all values of K: even at $K = 100$, Q4 contexts reach only 12.0% overlap compared to 39.2% for Q1.

Notably, model entropy does not increase in parallel with human entropy. Q4 contexts (highest human uncertainty) actually have the *lowest* mean model entropy (0.90 nats), while Q2 contexts show the highest (1.11 nats). The model remains confident regardless of whether human respondents agree or disagree.

Interpretation

The entropy-stratified analysis reveals a clear asymmetry between human and model uncertainty. When humans converge on a small set of likely completions (low-entropy contexts), the model's top predictions overlap substantially with the human response set. When human responses are highly

[1] Sums exceeding 1.0 arise because the model predictions include both partial prefixes and their completions as separate entries. For example, for the context targeting *банке*, both the prefix “бан” ($p = 0.996$) and the full continuation “банке” ($p = 0.996$) appear as distinct rows — their probabilities are per-path in the token tree, not per-leaf, so they are not mutually exclusive. The same pattern occurs in all 7 affected contexts (e.g., “покрыв”/“покрывало”, “раст”/“растительным”).

dispersed, the model fails to capture the long tail of plausible completions, and overlap drops sharply.

This dissociation is not simply a floor effect. Weighted overlap, which accounts for the probability mass of matched answers, shows the same gradient: the model captures 74.7% of the human probability mass in Q1 contexts but only 22.2% in Q4. This means the model is not merely missing rare answers in high-entropy contexts — it is also failing to predict the more probable responses.

The finding that model entropy is essentially flat (or even slightly *lower* in high-entropy contexts) suggests that GPT-4o-mini does not modulate its confidence in response to genuine predictive difficulty. Where humans hedge by distributing probability across many completions, the model maintains a concentrated distribution, producing a narrow beam of high-confidence predictions. This pattern is consistent with the known tendency of autoregressive language models to be overconfident and underestimate the entropy of natural language.

Target Word Analysis

Code: `analysis_target.py` · Data: `output/target/`

Method

To assess whether human and model predictions converge specifically on the *target* word (the word originally deleted from the sentence), we computed the probability mass each source allocates to the target lemma. For humans, $p_{\text{target}}^{\text{human}}$ equals the summed response proportion (`probability_y`) across all unique answers whose lemma matches the target lemma within a given context. For the model, $p_{\text{target}}^{\text{model}}$ equals the summed converted probability (`probability_converted`) across all unique predictions whose lemma matches the target, after deduplication by surface form (retaining the highest-probability variant).

We correlated $p_{\text{target}}^{\text{human}}$ and $p_{\text{target}}^{\text{model}}$ across all 144 contexts using both Pearson and Spearman coefficients. To examine how target-word convergence relates to broader distributional overlap, we divided contexts into four classes based on whether each source produced the target lemma at all ($p_{\text{target}} > 0$):

- **C1 — Both hit:** both humans and the model produced the target.
- **C2 — Only humans:** at least one human respondent produced the target, but the model did not.
- **C3 — Only model:** the model produced the target, but no human respondent did.
- **C4 — Neither:** neither source produced the target.

For each class we report mean p_{target}, lemma-based overlap@K, and probability-weighted overlap@K.

Results

The correlation between human and model probability mass on the target lemma was very strong: Pearson $r = 0.88$ ($p < 10^{-47}$) and Spearman $r = 0.76$ ($p < 10^{-27}$). When humans strongly agree on the target word, the model also tends to assign it high probability, and vice versa.

Table 1 shows the class breakdown. The largest class is C4 (86 of 144 contexts): in 60% of cases, neither humans nor the model produced the target word — these are contexts where the deleted word is not predictable from the left context alone. C1 contains 34 contexts where both sources hit the target; C2 has 22 where only humans did; C3 has just 2 where only the model did.

Class	N	Mean $p_{\text{target}}^{\text{human}}$	Mean $p_{\text{target}}^{\text{model}}$	overlap@10	w. overlap@10
C1 — both hit	34	0.384	0.378	0.335	0.622
C2 — only humans	22	0.096	0.000	0.146	0.332
C3 — only model	2	0.000	0.092	0.192	0.581
C4 — neither	86	0.000	0.000	0.193	0.359

Table 1: Target-word convergence classes: class sizes, mean target-word probabilities, and overlap metrics.

To illustrate each class, consider the following examples (Russian contexts shown with the target word in parentheses):

- **C1 — both hit:** *"Причиной аварии был мобильный"* (телефон). Humans assign $p = 0.95$, the model $p = 0.97$: a highly constraining context where both sources converge.
- **C2 — only humans:** *"Музыканты играли на похоронах, разгружали"* (вагоны). Humans produce the target ($p = 0.32$) alongside *обстановку* and *людей* (each $p = 0.11$), but it does not appear among model predictions at all — the context requires pragmatic world knowledge that the model lacks.
- **C3 — only model:** *"У тебя впереди замечательный день,"* (полный). No human respondent produced the target, yet the model ranks it second ($p = 0.12$) after the conjunction *и* ($p = 0.22$), followed by the near-synonym *наполненный* ($p = 0.07$). Only 2 contexts fall into this class.
- **C4 — neither:** *"В котёл бросают куски"* (баранины). The target word is not recoverable from the left context alone; both humans and the model distribute their probability mass across other continuations.

Interpretation

The strong correlation ($r = 0.88$) between human and model target-word probability confirms that GPT-4o-mini captures the contextual predictability of target words with high fidelity.

The class-based analysis reveals a clear pattern. In C1, where both sources converge on the target, weighted overlap@10 reaches 0.622 — indicating that shared success on the target word signals broader distributional alignment, not merely agreement on a single item. C2, where only humans hit the target (mean $p_{\text{target}}^{\text{human}} = 0.096$), shows the lowest overlap metrics (weighted overlap@10 = 0.332). The human probability mass in these contexts is small, suggesting marginal or morphological-variant hits that the model misses.

C3 contains only 2 contexts, too few for reliable generalization, but notably has high weighted overlap (0.581) despite zero human target probability — the model recovers the target in contexts where humans diverge to other answers.

The dominant class is C4 (86 contexts, 60%): neither source guesses the target, yet weighted overlap@10 is 0.359. This confirms that human–model distributional alignment does not depend on shared success at predicting the deleted word. Even when both fail on the target, they still converge on roughly a third of the probability mass over alternative continuations.